

Seth DeWhitt

Seattle, WA | sethjtdewhitt@gmail.com | sdewhitt.com | linkedin.com/in/seth-dewhitt | github.com/sdewhitt

EDUCATION

Purdue University, *West Lafayette, IN*

August 2023 – May 2027

B.S. in Computer Science, Software Engineering concentration

GPA: 3.6/4.0

Relevant Coursework: Data Structures, Analysis of Algorithms, Software Engineering, Systems Programming, Computer Architecture, C Programming, Software Testing, Object-Oriented Programming, Linear Algebra

Activities: Purdue Jazz Education Network (President), Purdue Jazz Band (Lead Tenor Saxophone)

Awards: Elmhurst University Jazz Festival Outstanding Soloist (2x), Dean's List (3x)

EXPERIENCE

Research Lead & Teaching Assistant

August 2025 – Present

The Data Mine, Purdue University | Git, Agile, Python, Typescript

West Lafayette, IN

- Directing 11 student researchers in collaboration with the Practical AI Podcast (PAI) and Prediction Guard
- Managing implementation of a RAG chatbot with Perplexity-style citations trained on episode content
- Architecting design of full-stack podcast generation app featuring voice clone models trained on PAI hosts
- Leading biweekly Agile sprints, coordinating tasks and progress reviews across a 50k+ line codebase

Software Engineering Intern

June 2025 – August 2025

Chewy | Python, React, Vite, FastAPI, Docker, AWS, Terraform, Jenkins, PostgreSQL

Bellevue, WA

- Streamlined data science workflows by provisioning dev environments on AWS ECS loaded with custom tooling
- Decreased admin overhead by **50%** and enhanced service scalability using AWS Fargate
- Automated distributed system maintenance with AWS Lambda and Kinesis, **lowering AWS expenses by 25%**
- Executed builds with Jenkins CI/CD pipelines, **decreasing deployment times by 66%**

Undergraduate Data Science Researcher | *Bayer Crop Sciences, Syngenta*

August 2023 – May 2025

The Data Mine, Purdue University | Python, R, Sci-kit learn, LLMs, Pandas

West Lafayette, IN

- Trained a LightGBM model on **500k+ rows** to model climate change impact on corn yields at **91% accuracy**
- Optimized model performance through feature engineering and hyperparameter tuning, reducing RMSE by **24%**
- Implemented data preprocessing and QC pipelines for **100M+ data points**, validated using mixed linear models

PROJECTS

MyPort | *TypeScript, OpenAI, LangChain, Supabase* | [GitHub](#) | [Link](#)

October 2025

- Pioneered RAG chatbot to answer portfolio questions via Supabase, OpenAI embeddings, and LangChain retrievers
- Built automated ingestion pipeline to chunk portfolio data and autosync, **supporting idempotent updates**

Evallm | *TypeScript, React, Next.js, Vercel, OAuth2, MongoDB* | [GitHub](#) | [Link](#)

July 2025 – August 2025

- Developed full-stack app using Groq API to evaluate user-prompted Meta LLM responses
- Implemented analytics dashboard with prompt analysis, LLM stats, and experiment tracking for model comparison
- Enhanced user experience with Llama real-time insights, Google authentication, and Tailwind-based UI

UNIX-like Shell | *C++, Lex/Yacc*

February 2025 – April 2025

- Developed a feature-rich command-line shell in C++ using Lex/Yacc for advanced parsing and execution.
- Implemented I/O redirection, pipes, subshells, signal handling (Ctrl-C, zombie elimination), and built-in commands
- Improved functionality with environment variable expansion, command history, and other interactive features

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C/C++, SQL, R

Cloud/Data: AWS, boto3, psycopg2, PostgreSQL, NoSQL, Pandas, Flyway, Supabase

Dev Tools: Git, FastAPI, React, Node.js, Next.js, Vite, Tailwind, Docker, Pinecone, MongoDB, Terraform, Jenkins